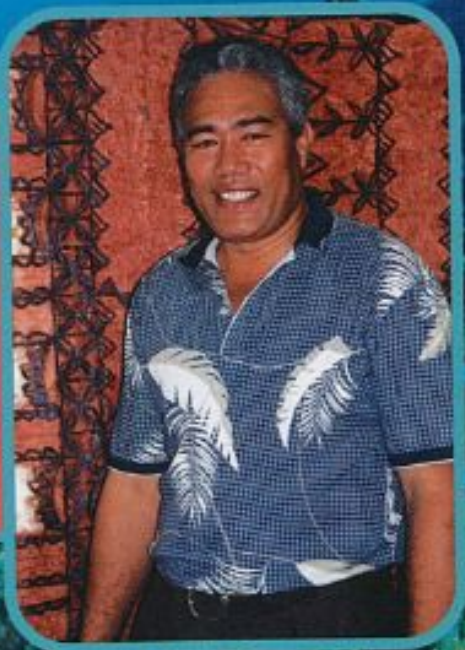
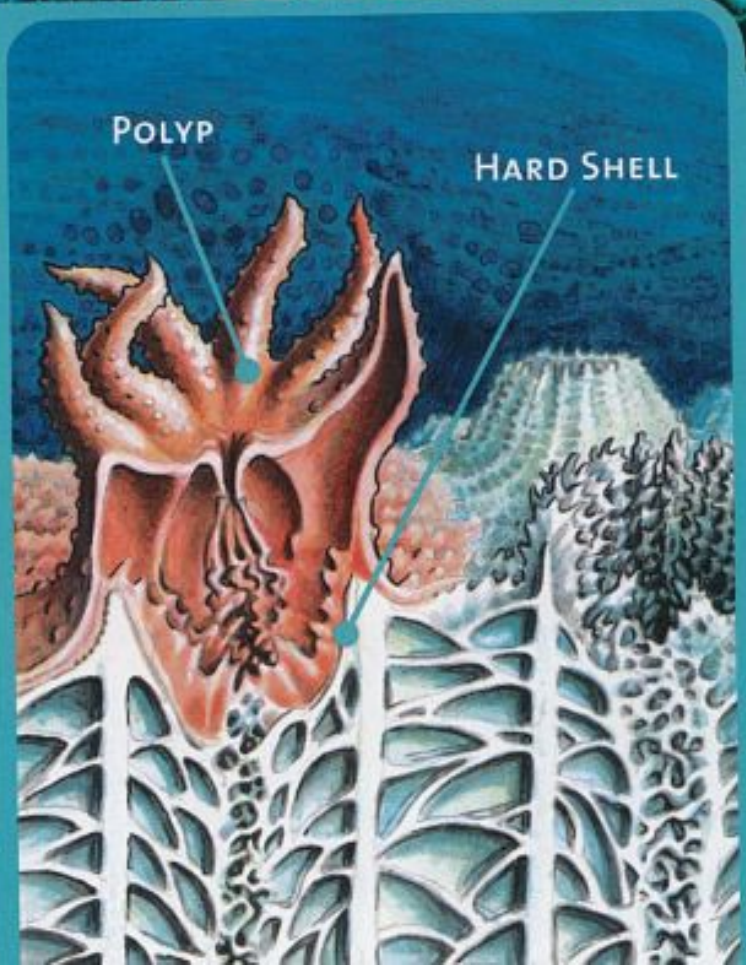
CORAL REEFS: AN ENDANGERED PARADISE

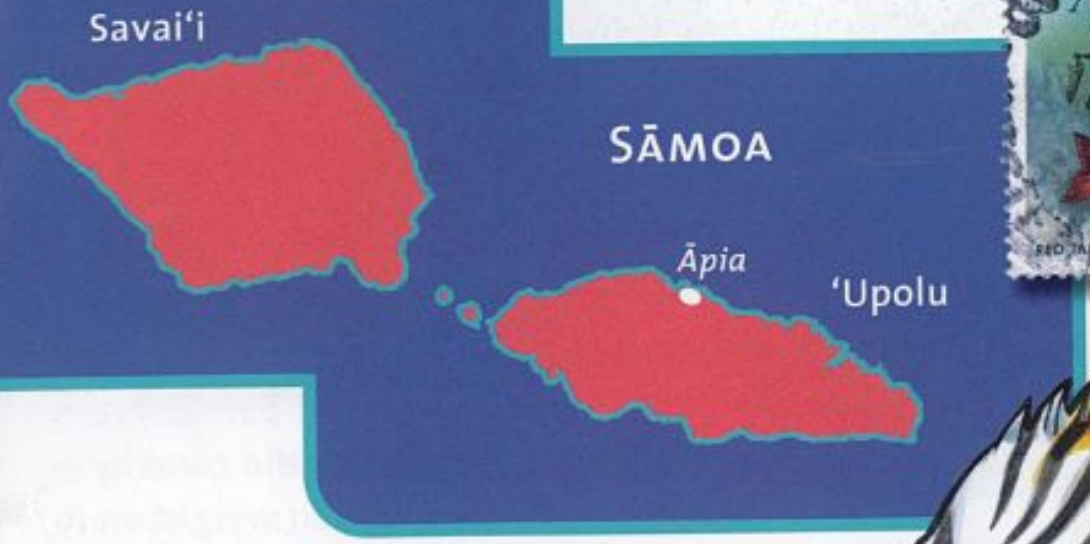
by Jill MacGregor

Mr Foua Toloa is a biologist who studies coral reefs in Sāmoa. He talks here to Jill MacGregor about this beautiful but threatened environment.



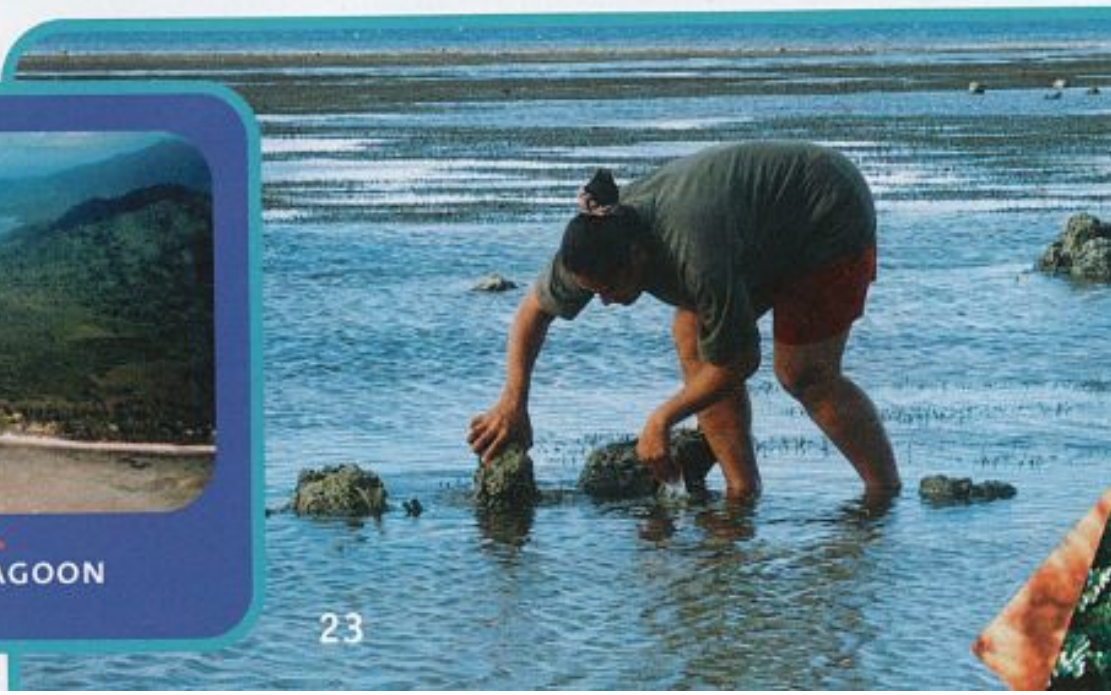


Coral reefs are living communities of coral polyps, which are tiny animals that are related to sea anemones. Each coral polyp builds a small shell around itself. When the tiny polyps die, their soft bodies rot away, but their hard shells are left behind. Soon, new polyps grow on top of the old shells. Over many years, this process creates very large natural sculptures of dead polyp shells. Only the very outer layer of the coral is actually alive.



When I was young, Sāmoa's coral reefs were clean and unspoiled. Night and day, the reef near our village was a busy place. Our whole family would go out onto the reef to fish and gather kaimoana. There was plenty of food, but we took only enough for our daily needs. At night, my father set eel traps. During the day, my mother collected shellfish and octopuses at low tide. On Saturdays, my friends and I went spear fishing or cast our nets to catch fish for the Sunday umu. Even the pigs used to snuffle around on the reef looking for food!

Nowadays, things have changed. Seafood is much harder to find. People are more likely to buy a tin of fish than to go out and catch some on the reef. A number of factors have caused this problem.



Coral Destruction

Over the years, people have destroyed a lot of coral. Villagers collected coral for decorating their gardens. Sometimes they used dynamite to blast off really big chunks. They carried this coral away in sacks, crushed it up, and used it for making paths and roads. Some of this "coral sand" was sold to concrete-making factories.

Tourists have bought huge numbers of coral souvenirs and have collected pieces of coral themselves. Sightseers can cause a lot of damage by walking on a reef at low tide. Scuba divers can harm the coral by touching it, resting their weight on it, or digging around in it.

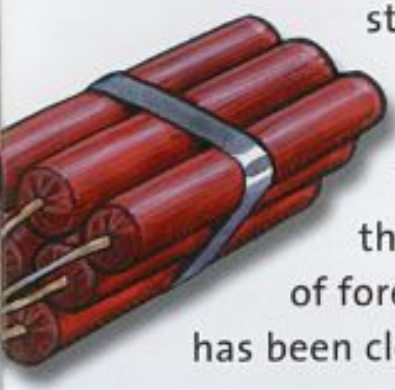
Natural events have also destroyed a lot of coral. In 1987, there was a population explosion of crown of thorns starfish. These starfish ate huge numbers of coral polyps. Many coral colonies died and began to crumble away.

To make things worse, cyclone Ofa struck in 1990 and was followed by cyclone Val in 1991. The cyclones caused gigantic waves, some of which were 25 metres high. The raging seas pounded the reefs of Sāmoa, ripping off massive chunks of coral.

STANHOPE ROAD
SCHOOL

Fishing Techniques and Pollution

Many people have used knives, hammers, and metal poles to prod among the coral while searching for sea creatures. Sometimes they've even blasted the reef with dynamite or poured bleach into the water to stun the fish, which then floated to the surface. These methods are now strictly banned.



Things that happen on the land often affect the sea as well. A lot of forest land in Sāmoa has been cleared for farming.

When land is cleared, some of the soil washes into rivers and is carried down into the sea. These fine soil particles are called silt. When the silt arrives in the still water of a lagoon, it sinks to the bottom and covers the coral. This kills the polyps.

SAVING THE REEFS

We're now very aware that our reefs are in trouble. Environmental groups around the country are making a big effort to save them. We're organising education programmes to tell the public about the problem. For example, Sāmoan schools run an annual beach clean-up day. This year, they organised a national school competition to design a logo for the Saving the Reef project. We now ask tourists not to collect or buy coral souvenirs.

My job is to organise workshops that educate villagers about the problem and its causes. With the community's help, a reef has a much better chance of returning to good health. Each village owns its reef and controls all fishing rights. A village may decide that only one half of a reef can be fished for a certain period of time. This takes the stress off the other half and allows it to recover.

Sāmoa is a very co-operative country. If one village is running low on shellfish in their lagoon, a nearby village may bring some round and set them free. In Āpia, we are setting up fish farms to increase the populations of at-risk species.

Hope for the Future

The results of reef recovery projects are recorded and sent to the Coral Growth Monitoring Programme, an international organisation. Villagers are very aware of their natural environment. They can therefore help scientists to keep track of local progress. Things are moving slowly, but we are seeing improvements that give us hope for the future.

A healthy coral reef is a beautiful environment with a stunning array of plants and animals. For thousands of years, our reefs have helped us to survive. Now it's time for us to return the favour!